

Homework 1: Pair-Clustering Model

Notes:

- It may be helpful to look at the materials of the block “03 Advanced features of multiTree” and of the tutorial paper by Schmidt, Erdfelder, & Heck (2023, *Psychological Methods*).
- It is sufficient to provide short bullet points and/or screenshots as answers.
- You can just copy output from multiTree without lengthy explanations or interpretations.

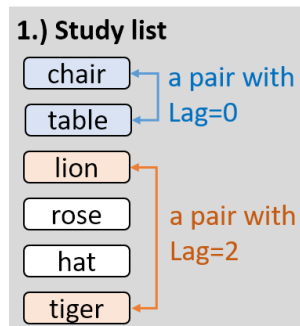
Exercises with multiTree

1. First, we apply the pair-clustering model to one group.
 - a) Specify the model equations in multiTree for word pairs (categories E1 to E4) and singletons (categories F1 and F2).
 - b) Provide the data of young participants only (see first table).
 - c) Fit the model and report model fit & parameter estimates.
2. Extend the model for two age groups (young vs. old adults).
 - a) Provide the extended set of frequencies (see first table).
 - b) Fit the model with separate parameters c , r , and u per group.
 - c) Test whether parameter r differs between the two age groups.
3. Extend and fit the pair-clustering model for the 2x2 design:
 - Factor A: young vs. old adults
 - Factor B: Lag = 0 vs. Lag = 15

$$\begin{aligned}
 p(E_1) &= c \cdot r \\
 p(E_2) &= (1 - c) \cdot u^2 \\
 p(E_3) &= (1 - c) \cdot 2 \cdot u \cdot (1 - u) \\
 p(E_4) &= c \cdot (1 - r) + (1 - c) \cdot (1 - u)^2 \\
 p(F_1) &= u \\
 p(F_2) &= 1 - u
 \end{aligned}$$

Category	young	old
E1	90	42
E2	14	5
E3	84	63
E4	212	290
F1	102	64
F2	298	336

Note that the lag between two words (Factor B) is only relevant for pairs (categories E1 to E4), but not for singletons (F1 and F2).



Category	young		old	
	Lag=0	Lag=15	Lag=0	Lag=15
E1	90	67	42	30
E2	14	18	5	13
E3	84	123	63	95
E4	212	192	290	262
F1	102		64	
F2	298		336	

Theoretical Questions

Above, we fitted the pair-clustering model to aggregated data for two age groups. The following theoretical questions refer to the block “Bayesian hierarchical MPT modeling”:

4. What are possible issues when using aggregated data?
5. Shortly explain the basic idea of a hierarchical MPT version of the pair-clustering model.
6. The latent-trait MPT approach assumes a covariance matrix Σ :
 - a) What dimensions does this matrix have for a pair-clustering model for a single group (as in exercise 1.)?
 - b) How can we interpret the values of this covariance matrix?